

HOW TO TEST FOR LL(1) GRAMMAR?

For every pair of production $A \rightarrow \alpha \mid \beta$

- $\text{first}(\alpha) \cap \text{first}(\beta) = \emptyset$
- At most one of α or β can derive empty string ϵ . And if $\beta \xRightarrow{*} \epsilon$ then
 - $\text{first}(\alpha) \cap \text{follow}(A) = \emptyset$

In LL(1) parsing, the **FOLLOW sets** of two variables can be interdependent, meaning the contents of one variable's FOLLOW set directly depend on the FOLLOW set of another.

Impact on LL(1) Parsing

This interdependence is crucial for constructing the **LL(1) Parsing Table**. Dependencies need to be propagated correctly else the Follow sets would be incomplete. An incomplete FOLLOW set means the parser won't know which lookahead tokens allow a variable to safely rewrite to ϵ , leading to parsing errors or an incorrect claim that a valid LL(1) grammar is not LL(1).

1. Compute $\text{first}()$ and $\text{follow}()$ sets for each non terminal of the grammar given below:
 - $S \rightarrow ABa \mid Bca$
 - $A \rightarrow cBCD \mid \epsilon$
 - $B \rightarrow Cda \mid ad$
 - $D \rightarrow bSf \mid a$
2. Test whether the following grammar is LL(1)
 - $S \rightarrow AB \mid cDa$

- $A \rightarrow ab \mid c$
- $B \rightarrow Dc$
- $C \rightarrow Ec \mid \epsilon$
- $D \rightarrow fD \mid \epsilon$

3. Is the following language LL(1)
 - $L = \{a^n cb^n \mid n \geq 1\}$
4. Design an LL(1) parse table for the following grammar
 - $S \rightarrow aAc d \mid BCe$
 - $A \Rightarrow b \mid \epsilon$
 - $B \rightarrow Cf \mid d$
 - $C \rightarrow fe$
5. Transform the following grammar so that it will be LL(1) without changing the language
 - $S \rightarrow aAC \mid bB$
 - $A \rightarrow Abc \mid Abd \mid e$
 - $B \rightarrow f \mid g$
 - $C \rightarrow h \mid i$

Hint : First left factor then remove left recursion

6. Construct LL(1) parse table for the following grammar:
 - $S \rightarrow aAC \mid bB$
 - $A \rightarrow eD$
 - $D \rightarrow bE \mid \epsilon$
 - $B \rightarrow f \mid g$
 - $C \rightarrow h \mid i$
 - $E \rightarrow eD \mid dD$